

Linear Circuits 2

Project #5 Circuit Design #2 Lab Report

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Objective

The goal of the project is to design a circuit that meets certain design requirements.

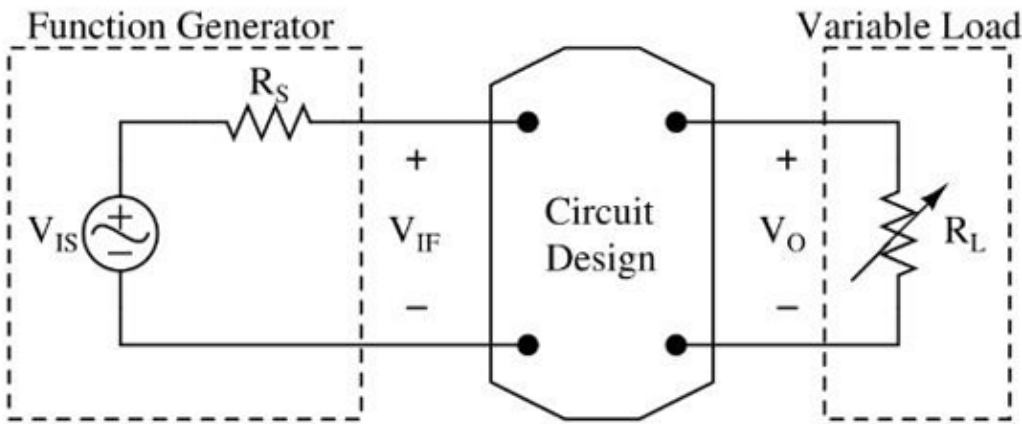


Figure 2 Circuit Model

Figure 1: Circuit model: function generator (V_{IS} , R_S) driving the circuit design under a variable load R_L

Basing our design off this circuit model, we are tasked to meet these requirements:

Design Specifications

$R_{MIN} = 1\text{ k}\Omega$
 $R_{MAX} = 30\text{ k}\Omega$

Frequency Range	$ V_O $
$1\text{ Hz} \leq f \leq 1\text{ kHz}$	$ V_O \leq 0.7 V_{IS} $
$10\text{ kHz} \leq f < 20\text{ kHz}$	$ V_O \geq 0.7 V_{IS} $
$f > 20\text{ kHz}$	$ V_O \geq 0.85 V_{IS} $

Notations

f Frequency
 $|V_{IS}|$ Input Signal Voltage Amplitude
 $|V_O|$ Output Voltage Amplitude

Figure 2: Design specifications: resistance range, frequency-dependent gain requirements, and notations

So not only must we meet all these specifications, but the design must also work for the entire range of load resistances.

Equipment

- Lab Equipment
- Circuit Simulator
- Hantek 3-in-1 Digital Equipment
- Powered Breadboard

Simulations / Hand Calculations

For all the simulations 10 V is used in a sinusoidal voltage source over varying frequencies. In addition to these, $50\ \Omega$ is added in series to replicate the function generator.

Initial Filter Behavior Simulation

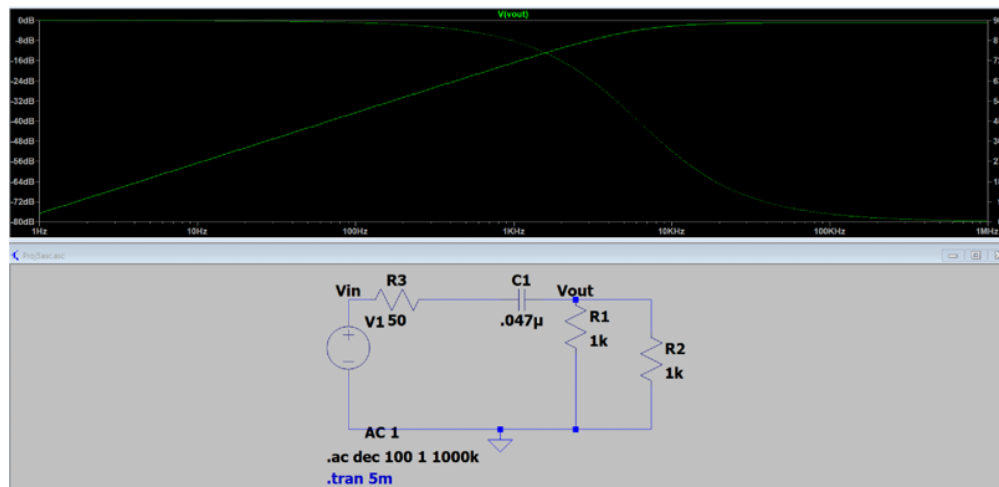


Figure 3: Frequency response and schematic with $R_2 = 1\ \text{k}\Omega$

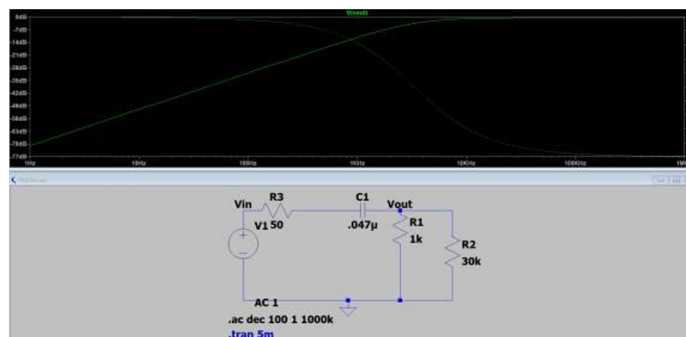


Figure 4: Frequency response and schematic with $R_2 = 30\ \text{k}\Omega$

Simulation 1) $R_L = 30\ \text{k}\Omega$, $f = 1\ \text{kHz}$, $C = 0.047\ \mu\text{F}$

Using these values in the circuit equates to $|V_O/V_{IS}|$ being 0.275.

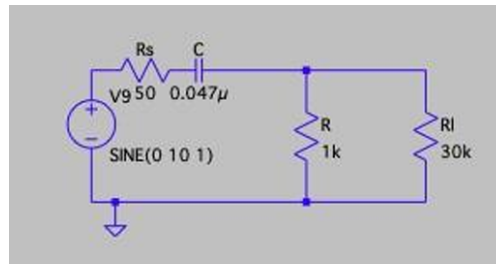


Figure 5: Simulation 1 schematic

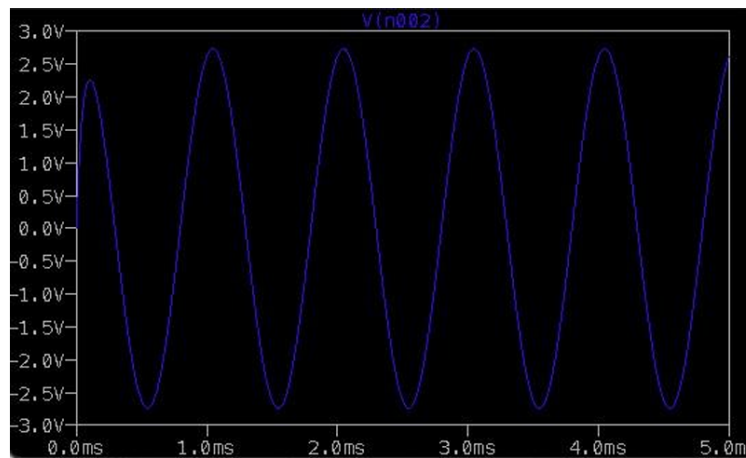


Figure 6: Simulation 1 output waveform

Above, is the simulated circuit and results of the gain, which align with the specification of $|V_O| \leq 0.7|V_{IS}|$.

Simulation 2) $R_L = 1\text{ k}\Omega$, $f = 10\text{ kHz}$, $C = 0.047\text{ }\mu\text{F}$

With these values the gain $|V_O/V_{IS}|$ of the circuit equates to 0.774.

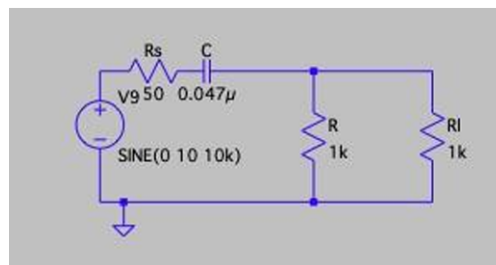


Figure 7: Simulation 2 schematic

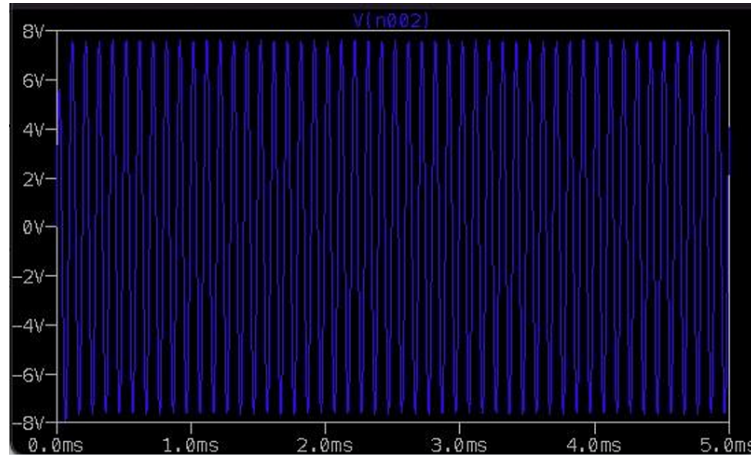


Figure 8: Simulation 2 output waveform over a 10 kHz span

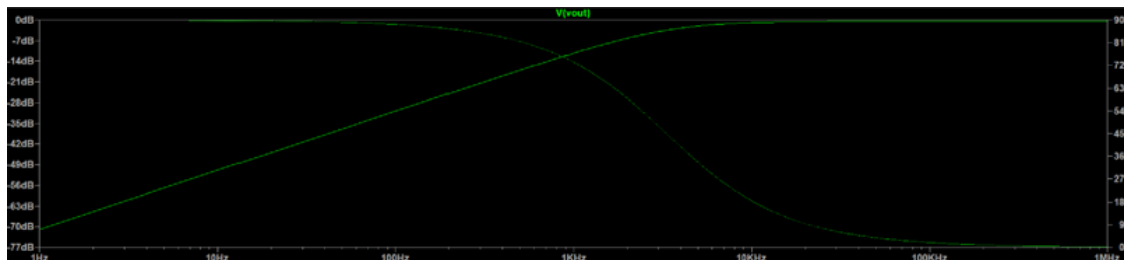


Figure 9: Simulation 2 frequency response

Once again, the pictures above demonstrate the simulated circuit and the waveform of V_O over the span 10 kHz. The gain of this circuit $|V_O/V_{IS}|$ equates to 0.774, which aligns with the specification of $|V_O| \geq 0.7|V_{IS}|$.

Simulation 3) $R_L = 1\text{ k}\Omega$, $f = 20\text{ kHz}$, $C = 0.047\text{ }\mu\text{F}$

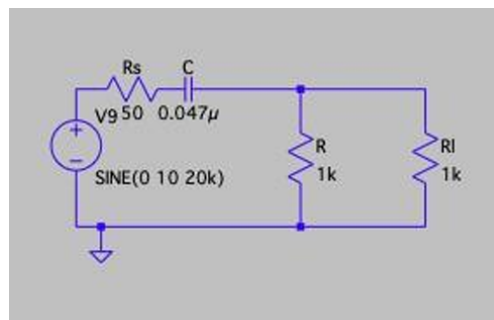


Figure 10: Simulation 3 schematic

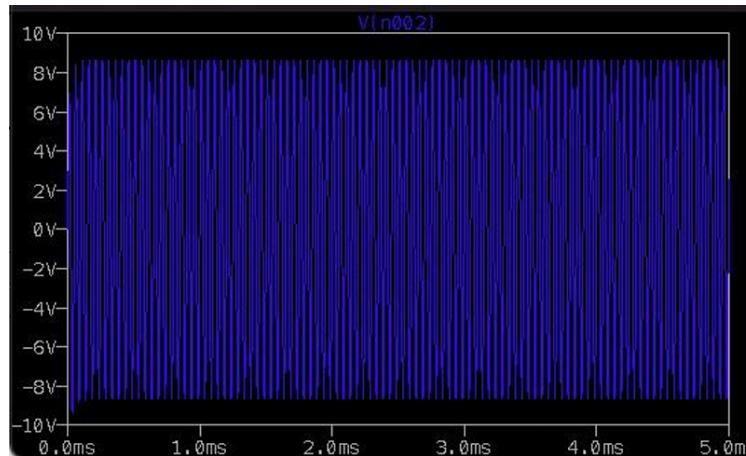


Figure 11: Simulation 3 output waveform over a 20 kHz span

This is the 3rd simulated circuit and its waveform over the span of 20 kHz. The gain of this circuit equates to 0.875, which follows the specification of this portion being $|V_O| \geq 0.85|V_{IS}|$.

Additional Hand Calculations

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$R_{eqn} = 11k\Omega$ $R_{max} = 30k\Omega$

Freq Range

$R_{eq} = R \parallel R_L$

$= 11k \parallel R_L$

If $R_L = 11k\Omega \Rightarrow R_{eq} = 5.5k\Omega$

$R_L = 30k\Omega \Rightarrow R_{eq} = 9.68k\Omega$

$$\left| \frac{V_O}{V_{IS}} \right| = \frac{R_{eq}}{(R_S + R_{eq})^2 + \left(\frac{1}{\omega C} \right)^2}$$

Figure 12: Hand calculations: equivalent resistance $R_{eq} = R \parallel R_L$ for the minimum and maximum load resistance

1) $|V_o| \leq 0.9 |V_{IS}|$ For $1\text{ Hz} \leq f \leq 1\text{ kHz}$

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi (1000)(49 \cdot 10^{-9})} = 3386 \Omega$$

$$\text{So } \left| \frac{V_o}{V_{IS}} \right| = \frac{968}{\sqrt{(50 + 968)^2 + 3386^2}} = 0.275$$

$$|V_o| = 0.275 |V_{IS}| \leq 0.9 |V_{IS}|$$

2) $|V_o| \geq 0.9 |V_{IS}|$ $10\text{ kHz} < f < 20\text{ kHz}$

$$R_{eq} = 500 \Omega \quad R_L = 110 \Omega$$

$$f = 10\text{ kHz} \quad X_C = \frac{1}{2\pi (10^4)(49 \cdot 10^{-9})} = 339 \Omega$$

$$\left| \frac{V_o}{V_{IS}} \right| = \frac{500}{\sqrt{(50 + 500)^2 + 339^2}} = 0.774$$

$$|V_o| \geq 0.9 |V_{IS}|$$

Figure 13: Hand calculations verifying the gain requirements for $1\text{ Hz} \leq f \leq 1\text{ kHz}$ and $10\text{ kHz} \leq f < 20\text{ kHz}$

Handwritten calculations on lined paper:

$$|V_o| \geq 0.85 |V_{ss}| \quad f > 20 \text{ kHz}$$

$f = 20 \text{ kHz}$

$$X_c = \frac{1}{2\pi (2 \cdot 10^4) (49 \cdot 10^{-9})} = 169 \Omega$$

$$\left| \frac{V_o}{V_{ss}} \right| = \frac{500}{\sqrt{550^2 + 169^2}} \approx 0.895$$

$$|V_o| \approx 0.85 |V_{ss}|$$

Figure 14: Hand calculations verifying the gain requirement for $f > 20 \text{ kHz}$

Experiment

To design the circuit, we knew it had to be a high pass filter, so we knew to use a capacitor and resistor. As far as the values, after calculation provided above, we concluded that a capacitor value of $0.047 \mu\text{F}$ and internal resistor of $1 \text{ k}\Omega$ would be sufficient to satisfy all requirements, with minimal difference on varying load resistances as well.

Bode Plot Behavior

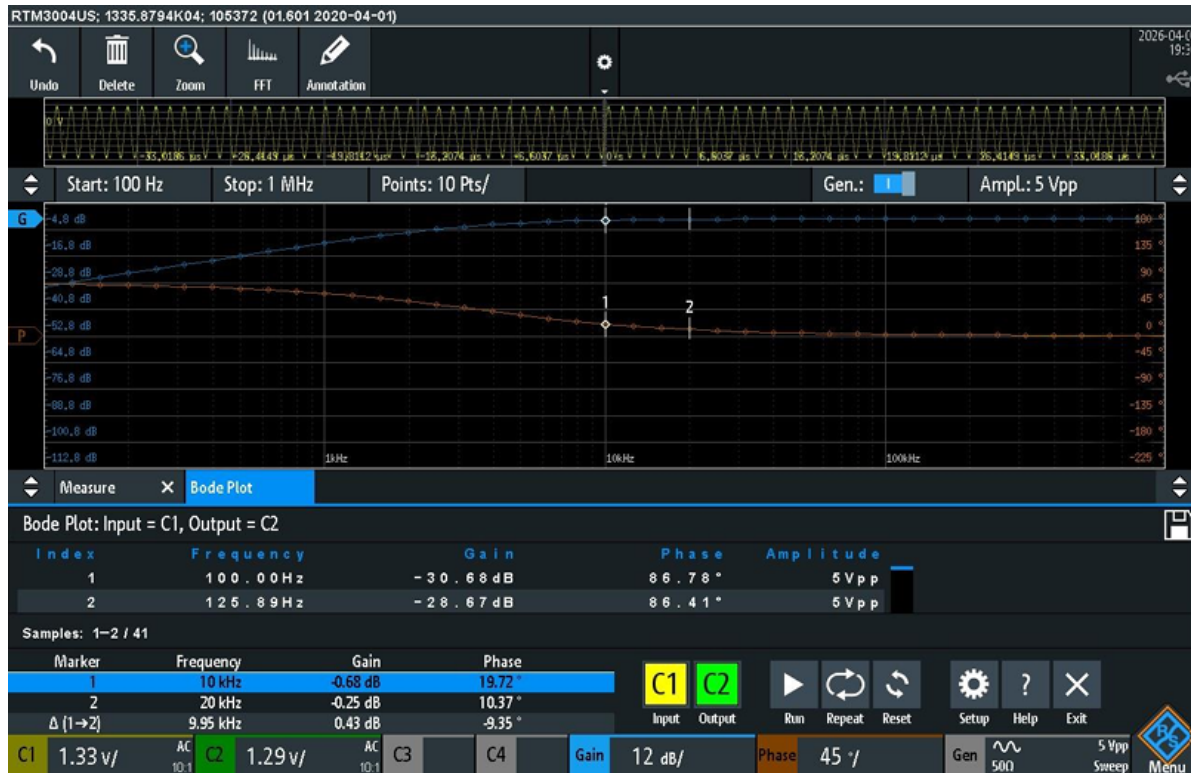


Figure 15: Measured Bode plot behavior of the filter

Scenario 1 Waveform: $R_L = 30\text{ k}\Omega$, $f = 1\text{ kHz}$

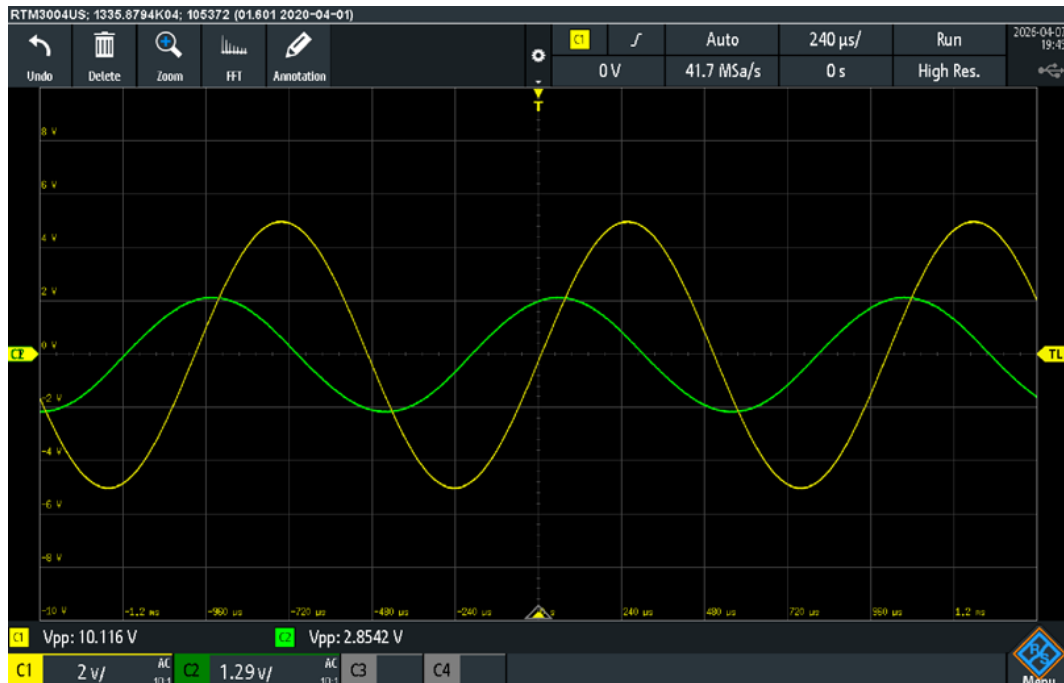


Figure 16: Scenario 1 measured waveform (C1: input, C2: output)

$|V_O/V_{IS}|$ with scenario 1's values were 0.275. In the picture above from the oscilloscope, C1 is the voltage input signal and C2 is the voltage output signal. So, comparing both signals it accurately reflects the initial requirement of $|V_O| \leq 0.7|V_{IS}|$.

Scenario 2 Waveform: $R_L = 1\text{ k}\Omega$, $f = 10\text{ kHz}$

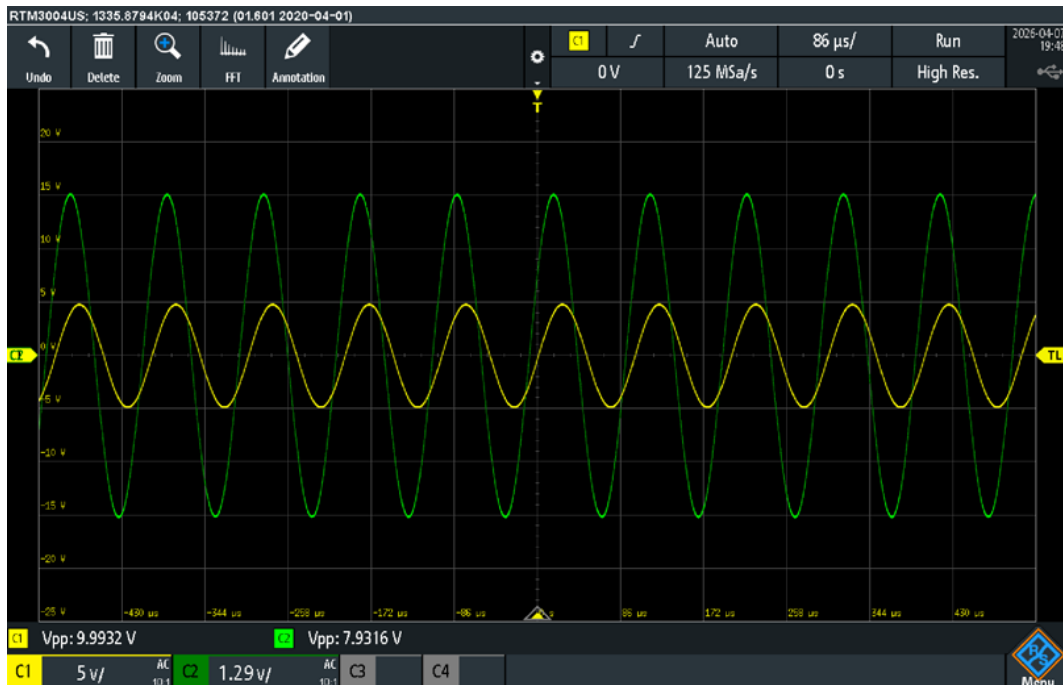


Figure 17: Scenario 2 measured waveform (C1: input, C2: output)

In scenario 2, $|V_O/V_{IS}| = 0.774$. This image from the oscilloscope has the same concept as the previous one, with Channel 1 being input and Channel 2 being output. After comparison and viewing the V_{pp} values you can see these waveforms satisfy the requirement of $|V_O| \geq 0.7|V_{IS}|$.

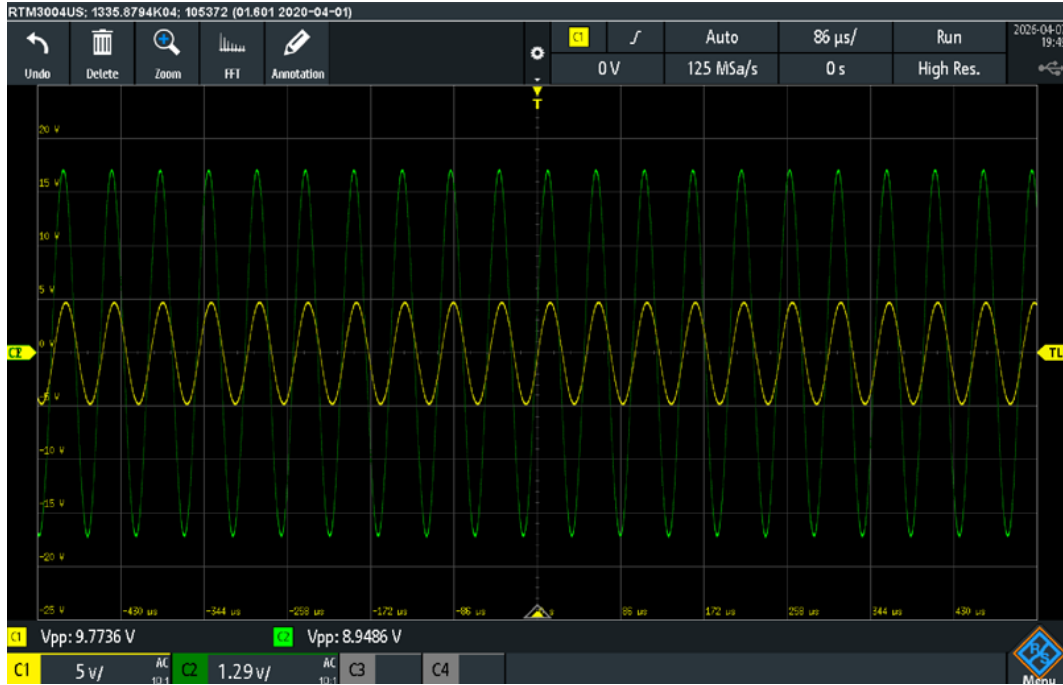
Scenario 3 Waveform: $R_L = 1\text{ k}\Omega$, $f = 20\text{ kHz}$ 

Figure 18: Scenario 3 measured waveform (C1: input, C2: output)

Lastly, in scenario 3, the $|V_O/V_{IS}| = 0.875$. From observing the oscilloscope, it shows the correct ratio of $|V_O| \geq 0.85|V_{IS}|$.

Results

Our output voltage matched all the constraints of the project, having values close to $0.875 V_{IS}$ for higher frequencies and close to $0.275 V_{IS}$ for lower frequencies. All in all, our filter circuit performed well, and the results are as expected.

Conclusion

Our circuit was a success, and we were able to use the properties of capacitors under high and low frequencies to make a filter under specific design constraints. We also tested the impact of the loading effect of the resistor in tandem with the internal resistance of the function generator and oscilloscope on filter response.